Project Progress Update

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Executive Summary:

This report outlines the mixed basket of products that will be included as the focus of our demand proxy, it also serves to further formalize some of the units of analysis, and shares our research into consumer segments for Berlin. The design follows the project blueprint's shift to weekly granularity as discussed, the established neighborhood-level analysis, and the next steps for refined development of a demand proxy based on our findings.

Key Assumptions:

- The four-item salad basket is representative of high-frequency organic fresh demand in Berlin.
- Weekly search behavior is directionally informative for near-term demand pressure.
- Neighborhood socio-demographic proxies are acceptable in absence of direct consumer panel data.

1. Product Basket Definition & Rationale

Following our last discussion in regard to the methodological blueprint, we began to consider potential products with which to focus our attention on. As such, we first identified potential product groupings to evaluate: salad essentials, root vegetables, cooking vegetables, herbs, and fruits.

Root Vegetables: Given that root vegetables (potatoes, carrots, onions) are essential staples we wanted to evaluate them as a potential basket. One finding from the research showed a strong willingness to pay for organics, largely driven by health consciousness, nutritional content, and reduced pesticide exposure (Grimm et al, 2022). However, with a longer shelf life in regard to

consumption frequency, as this durability reduces the frequency of customer purchases (Hu et al, 2024).

Fruits: Fruits (apples, berries, bananas) were considered due to their high visibility and consistent presence in the consumer shopping journey. These items are often driven by snacking or dessert-based behavioral triggers rather than meal preparation. While they represent a high-volume category, the extreme seasonality and volatility of certain items, specifically berries, create a consumption pattern that differs from the more stable, everyday produce items. Consequently, they appear to follow a distinct demand logic that suggests they may be better suited for a standalone category analysis.

Cooking Vegetables: Another category we identified was some popular cooking vegetables (zucchini, eggplant, broccoli) given that they are foundational to many household meals. Our assessment showed that these items are primarily purchased for cooked recipes, which separates them from the "raw-to-table" consumption occasions; however, this could lead to more “lumpy” demand patterns.

Herbs: Herbs and specialty greens were evaluated as they represent a rapidly evolving segment of the market. Products like kale, Thai basil, garden cress, and wasabi rocket have moved from niche items to daily staples in Berlin supermarkets, notably through Metro/Infarm partnerships, with some outlets now selling dozens of packs daily (Cointet et al, 2019). Despite this growth and the high-tech appeal of vertical farming, these items still represent a low overall total produce sales volume.

The **Salad Essentials** basket, comprising tomatoes, leafy greens, cucumbers, and bell peppers, was ultimately selected as the primary focus because it represents a highly cohesive

"raw-to-table" demand cluster. Unlike the excluded categories, these items share a high-frequency purchase cycle driven by their short shelf life and daily consumption habits.

Furthermore, this selection aligns with the strong consumer preference for organic produce in Berlin, where health consciousness and a desire for reduced pesticide exposure are primary drivers for "thin-skinned" vegetables.

Our mixed basket is defined as:

- Tomatoes (round/standard)
- Leafy greens (bagged mix)
- Cucumbers
- Bell peppers

These four products are treated as a behavioral cluster because they are jointly associated with one practical consumption grouping: quick, fresh, home-prepared salad meals. In demand terms, this creates a useful mix of:

- Complementarity (frequent co-purchase in salad occasions), and
- Substitution (e.g., cucumber vs pepper when relative prices move).

Kinds of produce:

Tomatoes (~8–12 varieties) Beef/Fleischtomate, cherry tomatoes (red, yellow, orange), cocktail tomatoes, Roma/San Marzano, vine tomatoes (Rispentomate), date tomatoes (Datteltomate), and occasionally heirloom varieties at better stores.

Cucumbers (~3–5 types) The classic Salatgurke (long, smooth), Snack-/Mini-gurken, Einlegegurken (pickling cucumbers), and sometimes Lebanese or Persian cucumbers at larger stores.

Leafy Greens (~10–15 varieties) Iceberg, Kopfsalat (butterhead), Romana/Cos, Lollo Rosso & Bionda, Feldsalat (lamb's lettuce — very popular in Germany), Rucola, Spinach, Mangold (Swiss chard), Chicorée, Radicchio, Pak Choi, Kale (Grünkohl), and mixed salad bags.

Bell Peppers (~4–6 types) Red, yellow, orange, and green are standard everywhere. Pointed peppers (Spitzpaprika) are also very common. Mini/snack peppers appear regularly too. True purple or white bell peppers are rare but occasionally spotted.

Stores:

- Variety expands significantly at **Edeka, Rewe, or Kaufland** vs. discounters like Aldi/Lidl
- **Turkish or Asian grocery stores** (very common in Berlin!) often carry 2–3x the variety
- Seasonal availability matters — summer brings far more options than winter
- Bio (organic) sections often stock rarer heritage varieties

Reasoning:

We chose these specific products because they are bought regularly by many customers (high volume from research) and are also items all of us in the team buy regularly. This provides us with intrinsic motivation to research about this bundle of produce more. Another reason is that these items are often eaten together, either in a salad and simply raw like that (very common in Germany).

1.1 Comparative Crop Characteristics

Dimension	Tomatoes (Round / Standard)	Leafy Greens (Bagged Mix)	Cucumbers	Bell Peppers
Primary role in basket	Anchor product	Core complement + convenience signal	Seasonal/weather-sensitive complement - May to August	Premium-tier complement
Typical perishability window	~5–7 days	~3–5 days	~4–7 days	~5–10 days
Seasonality profile	Year-round (greenhouse/import), seasonal quality effects	Increasingly year-round, including controlled-environment supply	Stronger warm-season sensitivity	Seasonal peaks + imports/greenhouse extension
Organic WTP hypothesis from	Relatively stronger WTP	WTP linked to freshness, health	Typically higher promotion sensitivity	Higher ticket, more sensitive to

research / elasticity pattern	for taste/quality in organic	framing, convenience		household budget pressure
Purchase frequency relevance	Weekly staple	High turnover, frequent replenishment	Weekly to bi-weekly depending on household	Weekly add-on, variable by recipe use

Interpretations for modeling:

- Greens carry high-frequency freshness behavior.
- Tomatoes capture broad organic intent with comparatively stable baseline demand.
- Cucumbers strengthen weather and seasonality signal detection.
- Peppers add premium-price sensitivity and substitution dynamics.
- Basket helps noise normalization as individual shocks (ie. price, shortage, recall, etc.) are smoothed

1.2 Further Berlin-Specific Considerations / Research

A. Vertical/Indoor Farming and Convenience Organic

Berlin-linked controlled-environment supply (especially for leafy categories and herbs) is relevant because it can reduce local volatility in freshness-sensitive products and support convenience-oriented channels (Specht et al, 2019). For modeling, this implies weaker pure weather dependence for part of the greens category than in open-field structures.

B. Seasonality vs Greenhouse/Import Circuits

Berlin demand is influenced by mixed sourcing: local-regional seasonal products, greenhouse supply, and import channels, including significant wholesale and ethnic-market pathways (Monaco et al, 2017). This reduces strict local seasonality and increases the importance of relative price and quality perception over origin alone.

C. Waste Reduction and Late-Day Markdown Practices

Retailers in Berlin increasingly apply markdown behavior to perishable inventory close to closing time, especially in produce-heavy assortments. This can create intra-week demand distortion (opportunistic purchases), which is another reason weekly aggregation is methodologically superior to daily noise-heavy demand tracing.

Three strategic realities must additionally be considered when shaping the framework:

1. Berlin's retail rhythm includes stock-up behavior due to Sunday closures in mainstream retail channels.
2. Germany's organic channel structure has shifted, with discounters becoming structurally important in organic sales and private-label penetration (BÖLW, 2025).
3. Organic demand in Berlin is culturally segmented, with strong heterogeneity across neighborhoods, income profiles, and food ideologies.

2. Unit of Analysis Specification

2.1 Temporal Unit: Weekly (Mon–Sun)

The model should use *weekly granularity* rather than daily because:

- shelf-life and pantry carry-over effects break daily interpretation,

- weekly windows absorb operational shocks,
- household purchase cadence in produce is predominantly weekly/bi-weekly.

2.2 Geographic Unit: Neighborhood Cluster Level

Demand should be modeled at a neighborhood-cluster level, with district-level proxies for:

- income,
- household composition,
- migration/ethnic retail ecosystems,
- organic retail density,
- convenience-delivery penetration.

Examples of analytically useful contrasts:

- Prenzlauer Berg / parts of Mitte: higher premium organic propensity,
- Neukölln/Kreuzberg sub-areas: stronger mixed-channel behavior including Turkish-market ecosystems,
- Outer districts (e.g., Lichtenberg segments): higher promotion sensitivity and larger basket-size optimization.

2.3 Weekly Composite Demand Index (0–100)

In focusing our evaluation on our basket selections, we start with the following setup:

Basket Demand Index (weekly) = weighted average of:

- Google Trends: "bio tomaten", "bio salat", "bio gurken", "bio paprika Berlin"
- Weighted by proposed product share

- Normalized to 0-100 scale
- Aggregated weekly (Mon-Sun)

Proposed Basket Demand Index

- 35% Tomatoes (highest volume, anchor product)
- 30% Leafy Greens (highest frequency, co-purchase driver)
- 20% Cucumbers (signal diversity, seasonal contrast)
- 15% Bell Peppers (premium tier, price elasticity data)

Practical note: This is a demand proxy, not observed sales. It should be calibrated against any available internal transaction or replenishment data as soon as accessible.

3. Structured Consumer Segments

3.1 Segment Profiles

1) Premium Sustainability Buyers

- Typical zones: Kreuzberg, Prenzlauer Berg, central high-income pockets.
- Core motive: quality, health, sustainability coherence.
- Behavior: low price elasticity, high response to product narrative (traceability, quality cues, wellbeing framing).
- Channel tendency: bio-specialists, curated supermarkets, premium delivery.

2) Price-Sensitive Discounters / Families

- Typical zones: mixed suburban and family-dense districts.
- Core motive: value optimization under budget constraints.

- Behavior: high promotion sensitivity; stronger shift toward discounter private-label organic after inflation pressure period.
- Channel tendency: Aldi/Lidl/Rewe price-led missions, larger planned baskets.

3) Convenience-Driven Urban

- Typical zones: dense inner-city rental/professional districts.
- Core motive: time efficiency, low-friction meal assembly.
- Behavior: moderate elasticity, high convenience premium tolerance if execution is fast/reliable.
- Channel tendency: quick-commerce, delivery-enabled chains, pre-packed mixes.

4) Community-Focused Traditionalists

- Typical zones: market-oriented neighborhoods, mixed-age communities.
- Core motive: trust, origin transparency, habitual purchase relationships.
- Behavior: medium elasticity; loyalty to known vendors and regional narratives.
- Channel tendency: weekly halls/markets, independent produce retail.

5) Modern Plant-Based Urbanites (Flexitarian-forward)

- Typical zones: urban younger cohorts across inner Berlin.
- Core motive: plant-forward diet identity without strict vegan exclusivity.
- Behavior: high responsiveness to social and recipe trends; relatively resilient organic demand if framed as culinary quality and not only ideology.
- Channel tendency: omnichannel, including social-influenced and delivery-heavy paths.

3.2 Interaction Model by Shock Type

Driver	Premium Sustainability	Price Sensitive Discounters / Families	Convenience-Driven Urban	Community-Focused Traditionalists	Plant-Based Urbanites
Price shock	Small demand drop; category stickiness	Largest demand drop/substitution	Medium response, basket simplification	Medium response, selective trade-down	Medium response, recipe substitution
Social media trend	Moderate if aligned with quality/sustainability	Low-moderate	High, especially convenience formats	Low	Very high
News/community sentiment	High response to safety/sustainability news	Moderate via media amplification	Moderate	High if trust channels affected	High for ethical narratives

Seasonality / holidays / events	Moderate premium seasonal uplift	Strong event-driven bulk shifts	Moderate, convenience-led	Strong around habitual market cycles	High around themed food moments
Weather signal	Moderate	Moderate (budget dominates)	Moderate-high for delivery substitution	Moderate via market attendance effects	High for fresh-meal planning behavior

4. Price Components

Component	Definition	Data Source
Own-price	Organic tomato price (EUR/kg)	AgriCom internal / Market Data
Substitute price	Conventional tomato price	AgriCom internal / Market Data
Complementary price	Prices of related basket items (lettuce, cucumber)	AgriCom internal / Market Data
Relative price gap	Organic premium vs conventional (%)	Calculated
Promotions	Temporary price reductions	AgriCom internal

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